Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/09/2007
PI Name:	Fritzsch, Bernd Ph.D.		
Project Title:	Neurobiology Of Vestibular Development In Mutant Mice		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:	NOTE (June 2016): Dr. Fritzsch is currently at University of Iowa		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	de Caprona, Dominique (Creighton University) Beisel, Kirk (Creighton University) Nichols, Dave (Creighton University)		
Grant/Contract No.:	NAG2-1595		
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Task Description:	To investigate the effect of normal and hypergravity on vestibular development in mice by investigating the effect of absence of otoconia on gravistatic behavior, neuronal connections and sensory epithelia gene expression. Using otoconia mutant mice we can investigate the absence of linear acceleration perception such as gravity on the behavior and connectional development of mice raised in 1g. Such genetically controlled manipulations allow for a more standardized analysis of a greater number of animals. By understanding the effect of absence of linear acceleration perception (gravity) on the functional development of ear input into the behavior and neuronal organization, we will be able to provide a more molecular understanding of the anticipated changes in the absence of gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our results from a study of embryonic rats developing in microgravity as well as several studies using hypergravity exposure suggest that exposure to altered gravity alters the development of gravistatic vestibular connections and causes long lasting behavioral changes. We now propose to examine how vestibular connections are affected by hypergravity and whether there is a critical age during which gravistatic vestibular connections must be exposed to 1g in order to develop normal connections. The connections of specific inner ear receptors to their central nuclei will be analyzed using the lipophilic dye, DiI, to trace fibers in aldehyde fixed tissue and to analyze their connections to vestibular neurons both light and electron microscopically. In a first, descriptive approach, we will examine the normal sequence of development and how it is altered by exposure to hypergravity by exposing developing rat embryos and pups to hypergravity (2g) for variable periods of time before and after birth. Subsequently we will seek a possible critical age during which the formation of abnormal connections between the ear and brain caused by hypergravity can be appropriately reorganized by intermittent exposure to 1g. For this we will expose rat pups to hypergravity with intermittent exposures to 1g to counteract the hypergravity-induced effects. It is hoped that this program will ultimately help to identify the ages during which biologically relevant gravity stimuli are essential for normal vestibular development. Thus it may be possible to minimize long-term effects of microgravity on the development of vestibular connections during long-term flights by using a 1g centrifuge at appropriate intervals.
Task Progress:	Progress on Aim1: Behavioral analysis of vestibular mutant mice. We have finalized data collection and have a paper in press on our behavioral data. This paper shows that mice without otoconia can not only learn behavioral balance tasks but do so with a steeper learning curve than their control littermates owing to the more profound handicap they are starting with. This paper refutes previous claims in the literature that removal of the ear eliminates spatial learning, likely because the sensory projection of the tlt-mutant mice is at least grossly normal whereas it is absent in animals that underwent ear removal. We have now extended this work to ask whether animals are able to move a 30o inclined bar up and down, both in normal light and in the dark. Surprisingly, the tlt mice with no otoconia performed extremely well in that behavioral task. We are currently finalizing these experiments in the no-cost extension period of this grant. If the first impression is supported by our statistics, we have to conclude that linear acceleration perception is for mice not relevant for simple balance tasks in the horizontal and inclined levels, a truly surprising finding. Title: Partial behavioral compensation is revealed in balance tasked mutant mice lacking otoconia. Authors: D. Crapon de Caprona, K.W. Beisel, D.H. Nichols & B.Fritzsch. Brain Research Bulletin, 64 (2004) 289-301. Abstract: We describe for the first time behavioral tests which show that mammals with congenital absence of otoconia can learn a motor task that normally relies on gravity perception. The mouse mutation tilted (tlt) occurs in the otopetrin 1 gene (Otop1tlt/tlt) and eliminates an essential component necessary for the formation of otoconia. Our data show that even in the absence of otoconia, tlt mutant mice, like normal mice, learn to cross a bar suspended between two boxes and, with practice, improve their speed of crossing. Despite this learned compensatory skills, tlt mutant mice show balance impairments, such as falling from the bar, not observed in wild type or heterozygous Otop1+/tlt littermates. The tlt mutant mice also use their tail as additional support, a behavior that is rarely exhibited in the control littermates. Interestingly, the Otop1+/tlt heterozygous littermates show in many aspects an intermediate phenotype between wild type and tlt mutant mice, suggestive of a gene dosage effect. Overall, these data support the notion that mammals can use other otic and extraotic receptors such as semicircular canals and limb proprioreceptors, respectively, to compensate for the absence of otoconia mediated gravity perception in a balance task. We have expanded this initial observation through the testing of movements down a 30o inclined bar. The general idea here is that in particular gravistatic input is essential for such movements. The evidence for this comes from initial observations on ear extirpated animals as well from the problems returning astronauts have going down stairs. The experiments have included a large set of mice but nevertheless show no statistical significant differences in overall performance of otoconia lacking mice (the tlt mutation). Most interesting were experiments in which we tested the effect of sudden light changes on trained animals while they were in the middle of the bar climbing down 60cm. As it turned out, tlt mice were completely unaffected by switching off the light. In contrast, some wildtype animals fell of the bar. We interpret this as suggesting that wildtype animals use a mixture of multiple inputs for this task, including vision. In contrast, tlt mutations seem to rely exclusively on proprioception and do not use vision in any way for the behavioral compensation. Combined these new set of observations will provide us with a roadmap for continued testing of the neuronal reorganizations underlying this apparent behavioral compensation. Progress in Aim 2: Development of connections of vestibular end organs to the brainstem and cerebellum. We have shown that the overall projection of tlt mutant mice is at least grossly normal (see the above cited paper) and are currently investigating the details of the saccular projections in neonates compared to the posterior crista. We concentrate on three areas, the projection to the cell group Y (which appears to be even more prominent in tlt mutant mice), the projection to the dorsal part of the medial vestibular nucleus (which appears to be reduced in tlt mutant mice) and the projection the uvula in the cerebellum (which appears to be less segregated than in control mice). We are currently analyzing more of these animals with longer diffusion times. To achieve better segregation of fiber staining we developed in collaboration with PTI Research modified lipophilic dyes. These new techniques have recently been published. *Fritzsch, B., Muirhead, K.A., Feng, F., Gray, B.D., (2005) Ohlsson-Wilhelm, B.M. (2005) Diffusion and imaging

	properties of three new lipophilic tracers, NeuroVue ™ Maroon, NeuroVue ™ Red and NeuroVue ™ Green and their use for double and triple labeling of neuronal profile. Brain Res. Bull., 66:249-258. [NOTE: see FY2006 report] More detailed analysis of the pattern of innervation of the vestibular organs from the CNS showed an unsuspected degree of segregation along the areas of different hair cell polarity. We have incorporated these new data into a manuscript in which we review the vestibular ear formation. *Beisel, K.W., Wang-Lundberg, Y., A. Maklad and Fritzsch, B. (2005) Development of the mammalian vestibular neurosensory ear. Vestibular Research, 15: 225-241.
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